How well do charmonium and bottomonium determine the B_c spectrum?

A potential-model test with the $B_c(2S)$ hyperfine difference

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October 19, 2026

We ask to what extent the spectrum of the B_c meson is fixed by charmonium and bottomonium data alone. Three families of spin-independent potentials—Cornell, logarithmic, and a power law with variable confining exponent α—are refitted, together with the quark masses and the strong coupling, to 27 J-resolved $c\bar{c}$ and $b\bar{b}$ levels from the 2026 Review of Particle Physics, with spin-dependent splittings treated in first-order perturbation theory. Without any B_c input the predictions for $B_c(1S)$, $B_c^*(1S)$ and $B_c(2S)$ agree across families to ± 15 MeV, while P-, D- and higher S-wave levels show 20–150 MeV model dependence that we quantify as a function of α. The $c\bar{c}+b\bar{b}$ data strongly favour a nearly logarithmic potential ($\alpha \lesssim 0.1$). The difference of hyperfine splittings $\Delta_{1S}-\Delta_{2S}$, which CMS and LHCb measure at 29–31 MeV, is predicted at 28–40 MeV and independently selects $\alpha \simeq 0.1$–0.2. With a fixed α_s the absolute $1S$ hyperfine splitting comes out at 76–82 MeV, well above the 64.5 ± 1.7 MeV recently measured by ATLAS; a one-loop running coupling with Λ determined by $c\bar{c}+b\bar{b}$ alone predicts $\Delta_{1S}=63$–72 MeV and $\Delta_{2S}=34$–38 MeV, reproducing the ATLAS, CMS and LHCb hyperfine measurements simultaneously without any B_c input. A full uncertainty budget (fit, data-set, scale and model dependence) gives ± 20–35 MeV on the excited levels and ± 15 MeV on the hyperfine splittings; a self-consistent smeared contact term with a mass-independent range is shown to be excluded by the $b\bar{b}$ hyperfine splitting. The spin-averaged $B_c(2S)$ is predicted 11–24 MeV above the measured value in every variant. We give masses, photon energies, $E1$ widths and radiative branching fractions for the $1P$, $2S$, $1D$ and $2P$ states as a guide to their observation, together with decay constants and explicit criteria that would falsify the description.

1 Introduction

The B_c meson is the only hadron built from two heavy quarks of different flavour. It is therefore both a laboratory for heavy-quark dynamics intermediate between charmonium and bottomonium and a stringent test of any description of the heavy quark–antiquark interaction: a model tuned on $c\bar{c}$ and $b\bar{b}$ must interpolate to $c\bar{b}$ with no additional freedom. Experimentally the field has moved quickly. The ground state is known to 0.3 MeV [1]; CMS [2] and LHCb [3] have observed the $B_c(2S)$ and the $B_c^*(2S)$ doublet; lattice QCD predicted the $B_c^*(1S)$ mass to better than 10 MeV [4]; and in 2026 ATLAS observed the $B_c^*(1S) \to B_c\gamma$ transition directly, measuring the hyperfine splitting $m(B_c^*)-m(B_c)=64.5\pm1.4\,^{+1.0}_{-1.4}$ MeV [5]. Excited P- and D-wave states are within reach of LHCb Run 3.

Non-relativistic potential models [6–8] have long been used to predict the B_c spectrum. Recent work has extended them to higher excitations and radiative decays [9, 10], and has begun to quantify parameter uncertainties with Bayesian methods [9]. A common feature of these studies, however, is that the model parameters are calibrated, at least in part, on the B_c states themselves. This makes agreement with $B_c(1S)$ and $B_c(2S)$ partly circular and obscures the question we wish to address here: *given only the $c\bar{c}$ and $b\bar{b}$ spectra, how well is the B_c spectrum determined, and which of its features are independent of the unknown form of the confining potential?*

We answer this by refitting three potential families to the current $c\bar{c}$ and $b\bar{b}$ data and propagating the fits to $c\bar{b}$ without further adjustment (Secs. 2–4). We then confront the hyperfine sector with three independent measurements: the ATLAS $1S$ splitting, and the $B_c(2S)$–$B_c^*(2S)$ doublet of CMS and LHCb, which determines the *difference* of the $1S$ and $2S$ splittings—a quantity in which the leading systematic uncertainties of the contact interaction cancel (Sec. 5). Radiative widths are given in Sec. 7 and we conclude in Sec. 9.

2 Framework

2.1 Spin-independent spectrum

For a quark of mass m_1 and antiquark of mass m_2 interacting through a central potential $V(r)$, the reduced radial wave function $u_{nl}(r)=rR_{nl}(r)$ obeys

$$u'' = 2\mu\Big[V(r)+\frac{l(l+1)}{2\mu r^2}-E_{nl}\Big]u, \qquad \mu=\frac{m_1m_2}{m_1+m_2}, \tag{1}$$

in natural units. We integrate Eq. (1) with the Numerov algorithm on a uniform grid ($h=10^{-3}\,\mathrm{GeV}^{-1}$) and locate eigenvalues by bisection on the number of radial nodes; the results are converged to better than 10^{-6} GeV with respect to h and the integration range. The spin-averaged mass is $M_{nl}=m_1+m_2+E_{nl}$.

We consider three families of potential:

$$V_{\mathrm{C}}(r) = -\frac{\kappa}{r}+\frac{r}{a^2}, \tag{2}$$

$$V_{\mathrm{L}}(r) = D + C\ln r, \tag{3}$$

$$V_{\alpha}(r) = V_0 + B\,(r^{\alpha}-1), \tag{4}$$

with r in GeV^{-1}. Eq. (2) is the Cornell potential [11]; Eq. (3) the Quigg–Rosner logarithmic potential [12]; Eq. (4) is the Martin power law [13] written so that the potential is pinned at $r=1\,\mathrm{GeV}^{-1}$ ($V_0=-1.166$ GeV) and the exponent α can be varied without shifting the

potential by amounts that the quark masses would otherwise absorb. For $\alpha = 0.1$ it coincides with the Martin potential of Ref. [6]; as $\alpha \to 0$ it tends to Eq. (3) with $C = B\alpha$; $\alpha = 1$ is linear confinement. The additive constants D and V_0 are exactly degenerate with $m_1 + m_2$ and are held fixed.

2.2 Spin-dependent splittings

Fine and hyperfine structure is added in first-order perturbation theory following Eichten and Quigg [6],

$$\Delta = \sum_{k=1}^{4} T_k, \tag{5}$$

$$T_1 = \Big[\frac{\langle \mathbf{L}\cdot\mathbf{s}_1\rangle}{2m_1^2} + \frac{\langle \mathbf{L}\cdot\mathbf{s}_2\rangle}{2m_2^2}\Big]\tilde{T}_1, \quad T_2 = \frac{\langle \mathbf{L}\cdot\mathbf{s}_1\rangle + \langle \mathbf{L}\cdot\mathbf{s}_2\rangle}{m_1 m_2}\tilde{T}_2,$$
$$T_3 = \frac{\langle \mathbf{s}_1\cdot\mathbf{s}_2\rangle}{m_1 m_2}\tilde{T}_3, \qquad T_4 = \frac{\langle S_{12}\rangle}{m_1 m_2}\tilde{T}_4, \tag{6}$$

with radial coefficients

$$\tilde{T}_1 = -\Big\langle \frac{1}{r}\frac{dV}{dr}\Big\rangle + 2\tilde{T}_2, \qquad \tilde{T}_2 = \frac{4\alpha_s}{3}\langle r^{-3}\rangle,$$
$$\tilde{T}_3 = \frac{8\alpha_s}{9}|R_{n0}(0)|^2, \qquad \tilde{T}_4 = \frac{\alpha_s}{3}\langle r^{-3}\rangle. \tag{7}$$

The full fitted potential is treated as a Lorentz scalar for the spin–orbit term, while the vector one-gluon exchange enters through α_s; this is the prescription of Ref. [6] and keeps the spin-dependent sector free of additional parameters. For $m_1 \neq m_2$ the states with $J = L$ mix; we construct the 2×2 matrix in the $\{{}^3L_L, {}^1L_L\}$ basis,

$$H = \begin{pmatrix} -\tfrac{1}{2}(a_1 + a_2) + 2d & \tfrac{1}{2}(a_1 - a_2)\sqrt{L(L+1)} \\ \tfrac{1}{2}(a_1 - a_2)\sqrt{L(L+1)} & 0 \end{pmatrix}, \tag{8}$$

with $a_i = \tilde{T}_1/2m_i^2 + \tilde{T}_2/m_1 m_2$ and $d = \tilde{T}_4/m_1 m_2$, and diagonalise it. Its trace and determinant agree with the jj-coupling matrices of Ref. [6], Eqs. (2.21). We label the two $J = L$ eigenstates L^+ and $L^{+\prime}$ in order of mass.

Two treatments of α_s are compared. In the *fixed* variant α_s is a single fitted constant. In the *running* variant we use the one-loop expression $\alpha_s(\mu) = 12\pi/[25\ln(\mu^2/\Lambda^2)]$ ($n_f = 4$) evaluated at a scale set by the quark masses, either $\mu = 2\mu_{\rm red} = 2m_1 m_2/(m_1 + m_2)$ or $\mu = \sqrt{m_1 m_2}$, and fit Λ instead of α_s. The two scale choices coincide for $c\bar{c}$ and $b\bar{b}$ and differ only in how the coupling is interpolated to $c\bar{b}$.

2.3 Radiative transitions

$E1$ and $M1$ widths are computed from the same wave functions [6],

$$\Gamma_{E1} = \frac{4\alpha\langle e_Q\rangle^2}{27}\, k^3\, (2J_f + 1)\, \mathcal{S}_{if}\, |\langle f|r|i\rangle|^2, \tag{9}$$
$$\Gamma_{M1} = \frac{16\alpha}{3}\, \mu_Q^2\, k^3\, (2J_f + 1)\, |\langle f|j_0(kr/2)|i\rangle|^2, \tag{10}$$

Table 1: Experimental input: 27 J-resolved levels below open-flavour threshold [1], in MeV. Our radial label n counts excitations from 1, so $\chi_c(1P)$ is $1\,{}^3P_J$. $\Upsilon(4S)$ is excluded.

System	State	Mass
$c\bar{c}$	$1\,{}^1S_0$	2984.1 ± 0.3
$c\bar{c}$	$1\,{}^3S_1$	3096.9 ± 0.0
$c\bar{c}$	$1\,{}^3P_0$	3415.5 ± 0.2
$c\bar{c}$	$1\,{}^3P_1$	3510.7 ± 0.1
$c\bar{c}$	$1\,{}^1P_1$	3525.4 ± 0.1
$c\bar{c}$	$1\,{}^3P_2$	3556.2 ± 0.1
$c\bar{c}$	$2\,{}^1S_0$	3637.8 ± 0.6
$c\bar{c}$	$2\,{}^3S_1$	3686.1 ± 0.0
$c\bar{c}$	$1\,{}^3D_1$	3773.7 ± 0.7
$c\bar{c}$	$1\,{}^3D_2$	3823.5 ± 0.3
$c\bar{c}$	$1\,{}^3D_3$	3842.7 ± 0.2
$b\bar{b}$	$1\,{}^1S_0$	9398.7 ± 2.0
$b\bar{b}$	$1\,{}^3S_1$	9460.4 ± 0.1
$b\bar{b}$	$1\,{}^3P_0$	9859.4 ± 0.5
$b\bar{b}$	$1\,{}^3P_1$	9892.8 ± 0.4
$b\bar{b}$	$1\,{}^1P_1$	9899.3 ± 0.8
$b\bar{b}$	$1\,{}^3P_2$	9912.2 ± 0.4
$b\bar{b}$	$2\,{}^1S_0$	9999.0 ± 3.5
$b\bar{b}$	$2\,{}^3S_1$	10023.4 ± 0.5
$b\bar{b}$	$1\,{}^3D_2$	10163.7 ± 1.4
$b\bar{b}$	$2\,{}^3P_0$	10232.5 ± 0.6
$b\bar{b}$	$2\,{}^3P_1$	10255.5 ± 0.6
$b\bar{b}$	$2\,{}^1P_1$	10259.8 ± 1.2
$b\bar{b}$	$2\,{}^3P_2$	10268.6 ± 0.6
$b\bar{b}$	$3\,{}^3S_1$	10355.1 ± 0.5
$b\bar{b}$	$3\,{}^3P_1$	10513.4 ± 0.7
$b\bar{b}$	$3\,{}^3P_2$	10524.0 ± 0.8

with $\langle e_Q\rangle = (m_b e_c - m_c e_{\bar{b}})/(m_b + m_c)$, $\mu_Q = (m_b e_c - m_c e_{\bar{b}})/4m_c m_b$, $k = (M_i^2 - M_f^2)/2M_i$, and $\mathcal{S}_{if} = 9\max(L_i, L_f)\{{}^{L_f\ J_f\ S}_{J_i\ L_i\ 1}\}^2$, which reduces to $\mathcal{S} = 1$ for ${}^3S_1 \leftrightarrow {}^3P_J$ and $\mathcal{S} = 3$ for ${}^1P_1 \leftrightarrow {}^1S_0$. For spin-mixed states the $E1$ rate is the sum over the $S = 1$ and $S = 0$ channels weighted by the squared mixing amplitudes. On charmonium this reproduces $\Gamma(\chi_{c0,1,2} \to J/\psi\gamma) = 163, 265, 340$ keV against the measured 151, 288, 374 keV [1].

All code is public (see Code and data availability) and every table and figure in this paper is regenerated by a single script.

3 Fit to charmonium and bottomonium

Table 1 lists the input. We fit each family's potential parameters, m_c, m_b and α_s (or Λ) by least squares to all 27 levels, using $\sigma^2 = \sigma_{\rm exp}^2 + \sigma_{\rm th}^2$ with a theory floor $\sigma_{\rm th} = 10$ MeV so that the keV-precision J/ψ and $\psi(2S)$ masses do not dominate a model whose intrinsic accuracy is tens of MeV. Parameter uncertainties are taken from the covariance of the fit and later propagated linearly to predictions.

The results are in Table 2 and Fig. 1. Residuals are typically 10–30 MeV and are largest for the

Table 2: Fitted parameters and χ^2 per degree of freedom for a 10 MeV theory floor. Units: B, C, m_c, m_b, Λ in GeV; a in GeV^{-1}. For the Martin family B, m_c and m_b are strongly correlated (see text); the individual errors are large but predicted masses are not.

Family	χ^2/dof	Parameters
Martin	84.7/22	$B = 7.19 \pm 0.45$, $\alpha = 0.096(6)$, $m_c = 1.63 \pm 0.25$, $m_b = 5.02 \pm 0.23$, $\alpha_s = 0.457(26)$
Log	73.3/23	$C = 0.737(5)$, $m_c = 1.509(4)$, $m_b = 4.908(2)$, $\alpha_s = 0.448(22)$
Cornell	133.7/22	$\kappa = 0.456(10)$, $a = 2.328(20)$, $m_c = 1.334(9)$, $m_b = 4.752(7)$, $\alpha_s = 0.373(19)$
Martin, running α_s	69.8/22	$B = 6.84 \pm 0.49$, $\alpha = 0.101(7)$, $m_c = 1.82 \pm 0.27$, $m_b = 5.20 \pm 0.26$, $\Lambda = 0.429(94)$
Log, running α_s	58.8/23	$C = 0.740(5)$, $m_c = 1.509(4)$, $m_b = 4.905(2)$, $\Lambda = 0.332(25)$

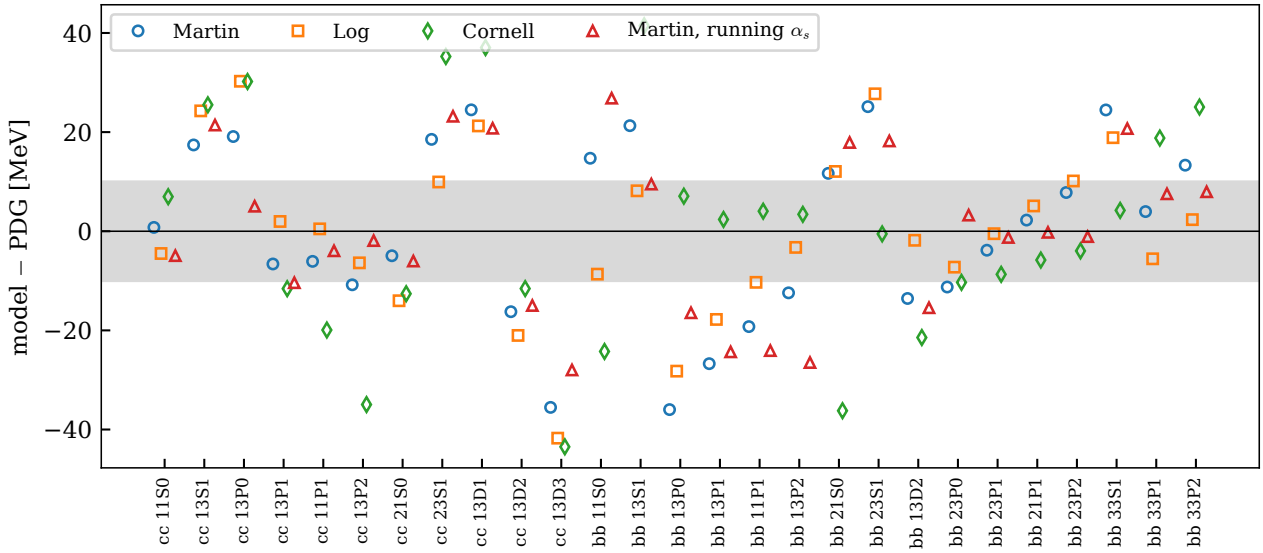


Figure 1: Residuals (model − PDG) of the fits for the three potential families and the running-coupling Martin fit. The grey band is the 10 MeV theory floor.

charmonium D waves (all families miss $\psi_3(3842)$ by $\sim$ 40 MeV), which lie at the $D\bar{D}$ threshold where coupled-channel effects are expected. Compared with the 1994 parameters [6], the logarithmic and Martin families change little except for α_s, which rises from 0.37–0.43 to 0.45–0.46; the Cornell quark masses, by contrast, drop by 0.4–0.5 GeV, confirming that the original Cornell masses were tuned to a different mass convention. The running coupling improves the fit in both families it was applied to (χ^2: 85 → 70 for Martin, 74 → 59 for logarithmic); the fitted $\Lambda = 0.33$–0.43 GeV corresponds to $\alpha_s(m_c) \simeq 0.50$ and $\alpha_s(m_b) \simeq 0.28$–$0.30$. The improvement comes mainly from the P-wave fine structure, which is sensitive to α_s through $\tilde{T}_2$ and $\tilde{T}_4$. Neither treatment reproduces both $1S$ hyperfine splittings within the theory floor: the fixed-α_s Martin fit gives 129 and 68 MeV for $c\bar{c}$ and $b\bar{b}$ (measured 113 and 62), the running-coupling fit 139 and 44 MeV. The first-order contact term, evaluated with the unperturbed wave function, is the least reliable part of the model, and we examine it further in Sec. 5.

For the Martin family the parameters B, m_c and m_b carry large formal errors because $B(r^\alpha - 1) \simeq B\alpha \ln r$ for small α, so B and the masses are nearly degenerate. The combination $m_c + m_b$ that enters B_c is nevertheless well constrained, and the propagated uncertainties on predicted masses are $\lesssim 15$ MeV.

To probe the shape of the confining potential we repeat the fit of Eq. (4) at fixed $\alpha \in \{0.05, 0.1, 0.2, 0.35, 0.5, 0.75, 1\}$ (Table 3, Fig. 2). The χ^2 rises monotonically from 73 at $\alpha = 0.05$ to 986 at $\alpha = 1$: the $c\bar{c}$ and $b\bar{b}$ spectra together strongly prefer a nearly logarithmic potential, consistent with the classic observation that the $2S$–$1S$ spacing is almost flavour independent [12]. A free fit of α returns 0.096(6).

4 The B_c spectrum

Table 4 gives the B_c spectrum through $3S$, $2P$ and $1D$ predicted by the three families, and Fig. 3 shows the dependence on α across the scan. Three features stand out.

First, the S-wave levels are remarkably stable. $B_c(1S)$, $B_c^*(1S)$ and $B_c(2S)$ vary by ±14, ±17 and ±6 MeV across the three families and by less than 70 MeV over the whole α scan, even though χ^2 changes by an order of magnitude. The ground state is reproduced at 6276 ± 14 MeV against the measured 6274.5 MeV without any B_c input.

Second, the excited P, D and $3S$ levels are not sta-

Table 3: Fits at fixed confining power α and the resulting B_c levels (MeV). Centroids are spin-weighted averages of the fine-structure multiplets. The last column is the predicted $\Delta_{1S}-\Delta_{2S}$; experiment gives 29.1 ± 1.7 MeV [2]. $\alpha=0$ denotes the logarithmic fit.

α	χ^2/dof	α_s	$B_c(1S)$	$B_c^*(1S)$	$B_c(2S)$	$1P$	$1D$	$3\,^3S_1$	$\Delta_{1S}-\Delta_{2S}$
0	73/23	0.448	6266	6349	6889	6768	7043	7253	40.3
0.05	72/23	0.461	6257	6337	6878	6752	7030	7254	37.2
0.1	85/22	0.457	6267	6343	6886	6753	7034	7270	33.8
0.2	136/23	0.465	6269	6338	6882	6737	7023	7285	27.3
0.35	254/23	0.465	6280	6340	6886	6724	7014	7312	19.4
0.5	404/23	0.462	6292	6344	6889	6713	7006	7336	13.1
0.75	689/23	0.457	6309	6351	6893	6698	6990	7371	5.3
1	986/23	0.451	6325	6360	6897	6686	6976	7400	0.0

Table 4: B_c spectrum (MeV) predicted from the $c\bar{c}+b\bar{b}$ fits with fixed α_s. Parenthesised errors are the propagated fit uncertainties; the model spread is half the range over the three families. The comparison columns are Eichten and Quigg (Buchmüller–Tye potential) [6] and PDG 2026 [1]. Our $1P$ and $1D$ correspond to the $2P$ and $3D$ of the principal-quantum-number labelling.

State	Martin	Log	Cornell	mean $\pm$ spread	EQ94	PDG
$1\,^1S_0$	6267(11)	6266(4)	6295(4)	6276 ± 14	6264	6274.5 ± 0.32
$1\,^3S_1$	6343(10)	6349(4)	6376(4)	6356 ± 17	6337	6339.0 ± 1.7
$1\,^3P_0$	6692(15)	6711(4)	6724(4)	6709 ± 16	6700	–
1^+	6735(13)	6753(3)	6753(3)	6747 ± 9	6730	–
$1^{+\prime}$	6756(10)	6770(3)	6759(2)	6762 ± 7	6736	–
$1\,^3P_2$	6774(10)	6787(3)	6770(3)	6777 ± 9	6747	–
$2\,^1S_0$	6886(11)	6889(2)	6877(3)	6884 ± 6	6856	6871.2 ± 1.0
$2\,^3S_1$	6928(10)	6931(2)	6930(3)	6930 ± 1	6899	–
$1\,^3D_1$	7029(14)	7040(3)	7036(3)	7035 ± 5	7012	–
$1D_2$	7029(9)	7037(3)	7022(3)	7029 ± 7	7009	–
$1D_2'$	7040(14)	7050(3)	7044(3)	7044 ± 5	7012	–
$1\,^3D_3$	7035(9)	7043(3)	7027(3)	7035 ± 8	7005	–
$2\,^3P_0$	7102(13)	7103(4)	7148(5)	7118 ± 23	7108	–
1^+	7132(12)	7131(3)	7176(4)	7146 ± 23	7135	–
$1^{+\prime}$	7151(10)	7147(3)	7185(4)	7161 ± 19	7142	–
$2\,^3P_2$	7164(10)	7158(3)	7196(4)	7173 ± 19	7153	–
$3\,^1S_0$	7240(11)	7225(4)	7290(6)	7251 ± 32	7244	–
$3\,^3S_1$	7270(10)	7253(4)	7335(6)	7286 ± 41	7280	–

ble: the $1P$ centroid moves by 80 MeV and the $3\,^3S_1$ by 150 MeV between $\alpha=0$ and $\alpha=1$, and the Cornell family—which the $c\bar{c}+b\bar{b}$ fit disfavours—sits 30–60 MeV above the others for the $2P$ and $3S$. These are the states for which future measurements will discriminate between confinement models; restricted to the $\alpha\lesssim0.2$ region preferred by $c\bar{c}+b\bar{b}$, the $1P$ centroid is 6737–6768 MeV and the $1D$ centroid 7023–7043 MeV.

Third, the spin-averaged $B_c(2S)$ is predicted 11–24 MeV *above* the measured 6871.2 MeV in every one of the thirteen fits performed in this work (Tables 4 and 3), with a propagated uncertainty of only 2–11 MeV. Since the $c\bar{c}$ and $b\bar{b}$ $2S$ levels are themselves reproduced to $\lesssim25$ MeV with residuals of both signs, this is a genuine interpolation failure at the few-MeV level rather than a fit artefact, and is the most robust discrepancy we find. We return to it in Sec. 9.

Our fixed-α_s $1P$ and $1D$ levels lie 5–30 MeV above those of Ref. [6], whose Buchmüller–Tye potential is softer at the relevant distances; the ordering within each multiplet and the mixing of the $J=L$ states (mixing angle $\simeq23^\circ$ for the $1P$) are the same.

5 The hyperfine test

In both the CMS and LHCb analyses the $B_c^*(2S)$ is reconstructed through $B_c^*(2S)\to B_c^*(1S)\pi\pi$, $B_c^*(1S)\to B_c\gamma$ with the soft photon unobserved, so the observed peak sits at $M(B_c^*(2S))-\Delta_{1S}$, where $\Delta_{nS}\equiv M(n^3S_1)-M(n^1S_0)$. The measured separation of the two peaks is therefore

$$M(B_c(2S))-M_{\rm peak}=\Delta_{1S}-\Delta_{2S}, \qquad (11)$$

$=29.1\pm1.5\pm0.7$ MeV (CMS [2]) and 30.9 MeV (LHCb [3]). This is a quantity in which potential models are unusually trustworthy. From Eq. (7),

$$\Delta_{1S}-\Delta_{2S}=\frac{8\alpha_s}{9m_cm_b}\left[|R_{10}(0)|^2-|R_{20}(0)|^2\right], \qquad (12)$$

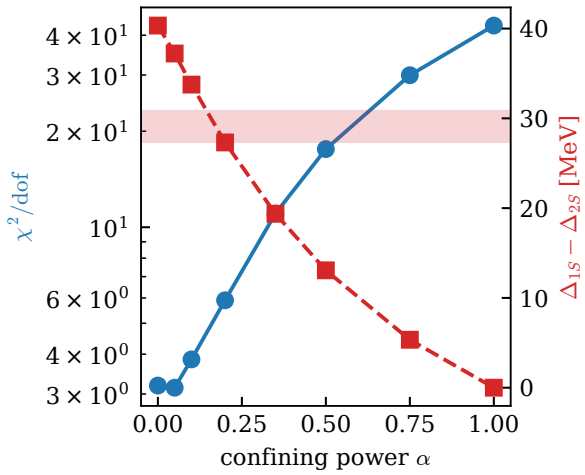


Figure 2: Left axis: χ^2/dof of the $c\bar{c}+b\bar{b}$ fit as a function of the confining power α in Eq. (4). Right axis: the predicted hyperfine difference $\Delta_{1S}-\Delta_{2S}$ of B_c; the band is the CMS measurement [2].

Table 5: Hyperfine splittings (MeV) of B_c predicted from $c\bar{c}+b\bar{b}$, for fixed and running α_s. Experiment: $\Delta_{1S}=64.5\pm1.7$ and $M(B_c^*)=6339.0\pm1.7$ (ATLAS [5]); $\Delta_{1S}-\Delta_{2S}=29.1\pm1.7$ (CMS [2]) and 30.9 (LHCb [3], statistical error ≈1.5); hence $\Delta_{2S}=35.4\pm2.4$. Lattice QCD (2010): $M(B_c^*)=6330\pm9$ [4].

Family	Δ_{1S}	Δ_{2S}	$\Delta_{1S}-\Delta_{2S}$	$M(B_c^*)$
Martin	75.8	42.0	33.8	6348
Log	82.4	42.0	40.3	6349
Cornell	81.1	53.0	28.1	6376
Martin, running ($\sqrt{m_c m_b}$)	62.8	35.0	27.8	6331
Martin, running (2μ)	67.3	37.5	29.8	6332
Log, running ($\sqrt{m_c m_b}$)	66.4	33.9	32.5	6342
Log, running (2μ)	72.0	36.8	35.3	6343

so the overall normalisation uncertainties of the contact interaction—the value of α_s, relativistic corrections to $|R(0)|^2$, the choice of quark masses—enter as a common factor, while the shape of the potential enters through the ratio $|R_{20}(0)|^2/|R_{10}(0)|^2$.

Table 5 and Fig. 4 show that the three fixed-α_s families predict $\Delta_{1S}-\Delta_{2S}=28$–$40$ MeV, bracketing the measurement, and that across the α scan (Fig. 2, Table 3) the prediction falls smoothly from 37 MeV at $\alpha=0.05$ to zero at $\alpha=1$, where $|R_{n0}(0)|^2$ is independent of n. The measurement therefore selects $\alpha\simeq0.1$–0.2, in agreement with—and independent of—the preference of the $c\bar{c}+b\bar{b}$ spectrum itself. Two unrelated observables point to the same confining potential.

The absolute splitting is a sharper test still, now that ATLAS has measured it directly: $\Delta_{1S}=64.5\pm1.7$ MeV [5], some 9 MeV (one standard deviation) above the 2010 lattice prediction [4]. Combined with the CMS difference this fixes $\Delta_{2S}=35.4\pm2.4$ MeV, so the two lowest hyperfine splittings of B_c are both known to a few MeV. Figure 5 and Table 5 compare them with the predictions. With fixed α_s all three families give $\Delta_{1S}=76$–82 MeV and $\Delta_{2S}=42$–53 MeV, 10–20 MeV too high, even though each reproduces the $c\bar{c}$ and $b\bar{b}$ $1S$ splittings to $\lesssim20$ MeV: a single coupling cannot interpolate the contact term $\propto\alpha_s|R(0)|^2/m_1m_2$ between flavours. With the one-loop running coupling the Martin family gives $\Delta_{1S}=63$–67 MeV, $\Delta_{2S}=35$–38 MeV and $M(B_c^*)=6331$–6332 MeV, and the logarithmic family $\Delta_{1S}=67$–72 MeV, $\Delta_{2S}=34$–37 MeV—both within or close to the ATLAS and CMS uncertainties. We stress that Λ in these fits is determined by the $c\bar{c}$ and $b\bar{b}$ data alone, and that neither a fixed coupling nor any adjustment of the spin-independent potential reproduces the measured B_c values. The agreement should nevertheless be read with the $c\bar{c}$ and $b\bar{b}$ residuals of Sec. 3 in mind: the running-coupling fit over-predicts the $c\bar{c}$ $1S$ splitting by 26 MeV and under-predicts the $b\bar{b}$ one by 17 MeV, and B_c lies between the two. What the result establishes is that the *interpolation* of the contact term to the intermediate reduced mass is accurate to a few MeV once the coupling runs, even though the endpoints are reproduced only at the 2σ level; with the uncertainty budget of Sec. 6 the prediction is $\Delta_{1S}=63\pm15$ MeV and $\Delta_{2S}=35\pm14$ MeV. The same fits place the lattice value [4] at the low edge of the allowed range; an updated lattice determination of $m(B_c^*)$ with current precision would be valuable.

Table 6: Fits with the hyperfine interaction included in the Hamiltonian as a Gaussian-smeared contact term of width σ (GeV; $\sigma=1$ unless stated), solved self-consistently for the S waves. Columns: χ^2/dof, fitted m_c (GeV), fitted $c\bar{c}$ and $b\bar{b}$ $1S$ splittings (measured 113 and 62 MeV), and the resulting B_c Δ_{1S}, Δ_{2S} (MeV; ATLAS/CMS: 64.5 ± 1.7, 35.4 ± 2.4).

Fit	χ^2/dof	m_c	$c\bar{c}$	$b\bar{b}$	Δ_{1S}	Δ_{2S}
Martin, fixed	111/22	1.04	108	18	37	14
Log, fixed	96/23	1.50	78	18	34	11
Martin, running	87/22	1.10	113	10	27	10
Log, running	68/23	1.50	91	11	27	9
Martin, running, $\sigma=0.8$	96/22	1.00	99	7	21	7
Martin, running, $\sigma=1.2$	82/22	1.25	119	13	32	12

5.1 A self-consistent contact term

Modern potential models usually include the spin–spin interaction in the Hamiltonian, regularised by a Gaussian of range $1/\sigma$, and solve for the singlet and triplet wave functions separately [14]; the first-order treatment above could in principle differ from this by tens of MeV. We have therefore repeated the fits with $V\to V+\langle\mathbf{s}_1\cdot\mathbf{s}_2\rangle\,\frac{32\pi\alpha_s}{9m_1m_2}\,\delta_\sigma(r)$, $\delta_\sigma(r)=(\sigma/\sqrt{\pi})^3e^{-\sigma^2r^2}$, for $\sigma=0.8$, 1.0 and 1.2 GeV, refitting α_s or Λ (Table 6). The outcome is unambiguous: with a mass-independent $\sigma\simeq1$ GeV the smeared term can reproduce the $c\bar{c}$ splitting but reduces the $b\bar{b}$ splitting to 7–18 MeV, because the compact $b\bar{b}$ wave function is averaged away over the smearing range, and the fit

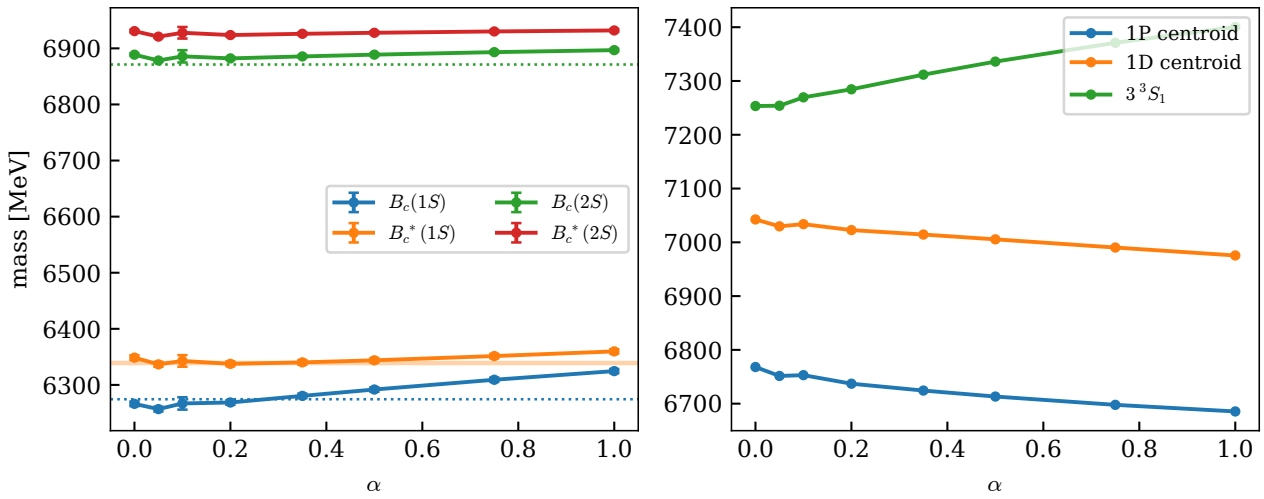


Figure 3: B_c levels as a function of the confining power α of the potential fitted to $c\bar{c} + b\bar{b}$. Left: S-wave states; dotted lines are the measured $B_c(1S)$ and $B_c(2S)$, the band is the ATLAS $B_c^*(1S)$ [5]. Right: $1P$ and $1D$ centroids and the $3\,^3S_1$.

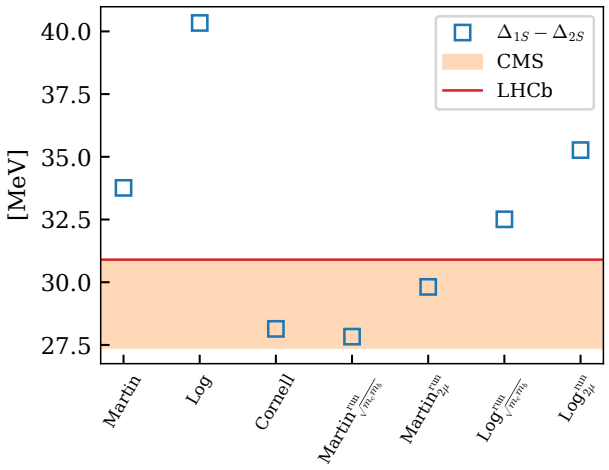


Figure 4: Predicted $\Delta_{1S} - \Delta_{2S}$ for each fit, compared with CMS (band) and LHCb (line).

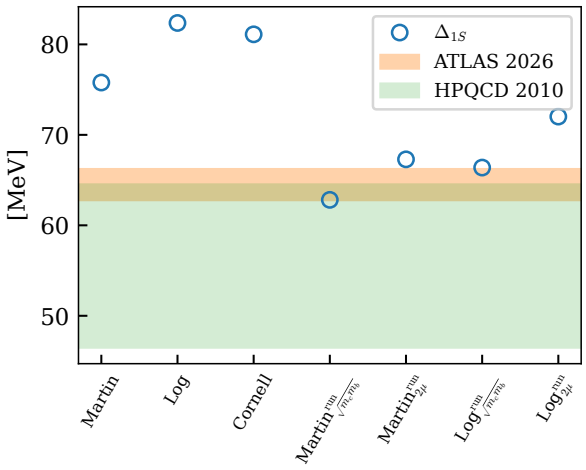


Figure 5: Predicted absolute $1S$ hyperfine splitting Δ_{1S} for each fit, compared with the ATLAS measurement [5] (narrow band) and the 2010 lattice prediction [4] (wide band).

compensates by pushing m_c to 1.0–1.25 GeV. The χ^2 is worse than in the first-order fits in every case, and the predicted B_c splittings (21–37 MeV) fall far below the ATLAS measurement. A smearing range that scales with the quark masses, as in the relativised model of Godfrey and Isgur, would be needed to cure this; we have not pursued it, since the first-order treatment already describes the data, but we note that the self-consistent and first-order treatments are not interchangeable at the present level of precision, and that the $b\bar{b}$ hyperfine splitting is the observable that discriminates between them.

6 Uncertainty budget and relativistic corrections

Table 7 gives the uncertainty budget. The dominant entry is the dependence on which levels enter the fit: removing the four charmonium D-wave states, which lie at the $D\bar{D}$ threshold and are the worst-described levels, or the two $b\bar{b}$ $3P$ states, moves the B_c predictions by 10–25 MeV. The theory floor, the α_s scale choice and the spread between the two running-coupling families are all small by comparison, and the propagated parameter covariance contributes 8–15 MeV. The totals are ± 25 MeV on the ground state, ± 20 MeV on the $2S$ doublet, ± 20–35 MeV on the $1P$ and $1D$ levels and ± 15 MeV on the hyperfine split-

Table 7: Uncertainty budget (MeV) for the B_c predictions of the running-coupling Martin fit. Sources: theory floor (5 and 20 MeV instead of 10; half range), data set (dropping the $c\bar{c}$ D waves or the $b\bar{b}$ $3P$ levels; largest shift), α_s scale ($\sqrt{m_c m_b}$ vs. 2μ), model (half range over the running-coupling Martin and logarithmic fits), and the propagated fit covariance. The total is the quadrature sum.

State	central	floor	data set	α_s scale	model	fit	total
$1\,^1S_0$	6268	0	22	3	4	10	25
$1\,^3S_1$	6331	0	19	1	6	8	22
$2\,^1S_0$	6884	0	15	2	5	10	19
$2\,^3S_1$	6919	0	13	1	5	9	17
$1\,^3P_0$	6699	0	25	6	16	15	34
1^+	6734	0	21	3	14	13	29
$1^{+\prime}$	6746	0	16	1	11	9	21
$1\,^3P_2$	6760	0	14	2	9	7	19
$1\,^3D_1$	7025	0	18	2	11	13	25
$1\,^3D_3$	7026	0	11	1	7	8	15
$3\,^3S_1$	7263	0	7	0	5	9	12
Δ_{1S}	63	0	4	4	2	13	15
Δ_{2S}	35	0	2	2	1	13	14

Table 8: Pseudoscalar decay constants (MeV) from the running-coupling Martin fit, $f^2 = 3|R(0)|^2 C/\pi M$ with the one-loop correction $C = 1 - \frac{\alpha_s}{\pi}\left[2 - \frac{m_1 - m_2}{m_1 + m_2} \ln \frac{m_1}{m_2}\right]$, compared with lattice QCD [15, 16].

Meson	this work	lattice
η_c	455	395(2)
η_b	603	667(6)
B_c	465	427(6)

tings; the latter are limited by the parameter covariance, i.e. by the precision with which the $c\bar{c}$ and $b\bar{b}$ data fix Λ.

The size of the neglected relativistic corrections is indicated by $\langle v^2 \rangle = \langle p^2 \rangle / m^2$ for each quark, which we obtain from the solved wave functions through $\langle p^2 \rangle / 2\mu = E - \langle V_{\rm eff} \rangle$. For the $1S$ states $\langle v_c^2 \rangle = 0.20$ in charmonium, $\langle v_b^2 \rangle = 0.07$ in bottomonium, and in B_c $\langle v_c^2 \rangle = 0.29$, $\langle v_b^2 \rangle = 0.035$; the P and D waves are less relativistic (0.12 and 0.08 for the charm quark in B_c), the $2S$ and $3S$ slightly more (0.31, 0.32). The charm quark in B_c is thus *more* relativistic than in charmonium, because the heavy b antiquark pulls the wave function inwards. Relativistic corrections of order $\langle v^2 \rangle$ to the kinetic energy are largely absorbed into the fitted potential and quark masses, as they are for $c\bar{c}$; the residual effect on B_c is a difference of order $(\langle v_c^2 \rangle_{B_c} - \langle v_c^2 \rangle_{c\bar{c}}) \times$ (a few hundred MeV), i.e. 10–30 MeV, which is consistent with the size of the $B_c(2S)$ discrepancy and with the ± 25 MeV budget.

The same $|R(0)|^2$ that drives the hyperfine splitting determines the leptonic decay constants (Table 8). They agree with lattice QCD to 10–15% with the characteristic pattern of potential models—too large for $c\bar{c}$ and B_c, too small for $b\bar{b}$—which is the same pattern seen in the $1S$ hyperfine residuals of Sec. 3 and confirms that it originates in $|R(0)|^2$ rather than in the spin-dependent formalism.

7 Radiative transitions

Table 9 lists the $E1$ and $M1$ widths of the $1P$, $1D$ and $2P$ states. The $1P \to 1S$ widths of 60–135 keV agree in magnitude with Refs. [6, 10]; the spin mixing of the 1^+ states redistributes the rate so that the heavier $1^{+\prime}$ decays mainly to $B_c\gamma$ (135 keV) and the lighter 1^+ mainly to $B_c^*\gamma$ (71 keV). The $1D \to 1P$ widths of 40–95 keV make the cascade $1D \to 1P\gamma \to 1S\gamma\gamma$ the natural route to the first D-wave B_c state. The $M1$ rates are $\lesssim 0.1$ keV; the $B_c^*(1S) \to B_c\gamma$ photon ($k \simeq 60$–75 MeV) remains the principal obstacle to a direct B_c^* mass measurement. We have not included the model dependence of these widths, which is dominated by the photon energy ($\propto k^3$) and is therefore largest for transitions involving the excited levels of Sec. 4.

8 Predictions for experiment

Table 10 collects the predictions in the form most useful for a search. Several features are robust across all our fits and are worth stating explicitly.

The $1P$ multiplet. Four states between 6699 and 6760 MeV, all below the BD threshold and therefore narrow, decaying radiatively to the $1S$ doublet with photons of 360–460 MeV. The two $J^P = 1^+$ states are separable by their final state: the lighter, 1^+, is 85% $^3P_1$ and decays to $B_c^*\gamma$ (391 MeV photon, 88%); the heavier, $1^{+\prime}$, is 85% $^1P_1$ and decays to $B_c\gamma$ (461 MeV, 94%). Since the $B_c^* \to B_c\gamma$ photon is only 64 MeV and is rarely reconstructed, the 1^+, $^3P_0$ and $^3P_2$ will appear as peaks in $B_c\gamma$ at masses shifted down by 64 MeV, exactly as the $B_c^*(2S)$ does; the $1^{+\prime}$ will appear at its true mass. The predicted $^3P_2 - ^3P_0$ splitting is 46–82 MeV over the fitted families.

Table 9: $E1$ widths above 10 keV and $M1$ widths of the lowest B_c states, for the Martin fit with fixed α_s. k is the photon energy. Spin-mixed states are treated as described in Sec. 2.

Transition	Type	k [MeV]	Γ [keV]
$1^{+\prime} \to 1\,^1S_0\,\gamma$	E1	472	135.3
$1\,^3P_2 \to 1\,^3S_1\,\gamma$	E1	417	110.8
$1\,^3D_1 \to 1\,^3P_0\,\gamma$	E1	329	94.6
$1\,^3D_3 \to 1\,^3P_2\,\gamma$	E1	256	80.6
$1^+ \to 1\,^3S_1\,\gamma$	E1	381	71.4
$1^{+\prime} \to 2\,^1S_0\,\gamma$	E1	260	65.7
$1\,^3P_0 \to 1\,^3S_1\,\gamma$	E1	340	60.1
$2\,^3P_2 \to 2\,^3S_1\,\gamma$	E1	232	57.1
$1D_2' \to 1^+\,\gamma$	E1	298	52.5
$1D_2 \to 1^{+\prime}\,\gamma$	E1	267	46.7
$1D_2' \to 1^{+\prime}\,\gamma$	E1	278	46.2
$1D_2 \to 1^+\,\gamma$	E1	287	42.4
$1\,^3D_1 \to 1^+\,\gamma$	E1	288	40.2
$1^+ \to 2\,^3S_1\,\gamma$	E1	202	30.4
$2\,^3P_0 \to 2\,^3S_1\,\gamma$	E1	172	23.1
$1^+ \to 1\,^1S_0\,\gamma$	E1	452	21.5
$1^{+\prime} \to 1\,^1S_0\,\gamma$	E1	829	18.7
$2\,^3P_2 \to 1\,^3S_1\,\gamma$	E1	774	18.6
$1^{+\prime} \to 1\,^3S_1\,\gamma$	E1	401	15.0
$2\,^3S_1 \to 1\,^3P_2\,\gamma$	E1	152	15.0
$2\,^3P_0 \to 1\,^3S_1\,\gamma$	E1	718	14.9
$2\,^3S_1 \to 1^+\,\gamma$	E1	190	14.7
$2\,^1S_0 \to 1^{+\prime}\,\gamma$	E1	128	13.7
$1^+ \to 1\,^3S_1\,\gamma$	E1	746	13.6
$2\,^3P_2 \to 1\,^3D_3\,\gamma$	E1	128	12.8
$1^+ \to 2\,^1S_0\,\gamma$	E1	242	11.9
$1D_2' \to 1\,^3P_2\,\gamma$	E1	261	11.5
$2\,^3S_1 \to 1\,^3P_0\,\gamma$	E1	232	10.5
$1\,^3S_1 \to 1\,^1S_0\,\gamma$	M1	75	0.1
$2\,^3S_1 \to 1\,^1S_0\,\gamma$	M1	629	0.1
$2\,^1S_0 \to 1\,^3S_1\,\gamma$	M1	522	0.1
$2\,^3S_1 \to 2\,^1S_0\,\gamma$	M1	42	0.0

The $2S$ doublet. $B_c(2S) = 6884 \pm 19$ MeV and $B_c^*(2S) = 6919 \pm 17$ MeV, a splitting of 35 ± 14 MeV. The measured $B_c(2S)$ is 13 MeV below our central value, at the edge of the budget; a measurement of the $B_c^*(2S)$ mass through the reconstructed $B_c^*(1S)$ photon, now feasible after Ref. [5], would test Δ_{2S} directly.

The $1D$ multiplet. Four states within 15 MeV of each other at 7020–7035 MeV, well below the BD threshold ($\approx$ 7145 MeV) and therefore also narrow. They cascade to the ground state through the $1P$ levels with two or three photons of total energy 730–750 MeV; the $^3D_3 \to {}^3P_2\gamma$ (261 MeV) and $^3D_1 \to {}^3P_0\gamma$ (319 MeV) lines are the cleanest. The spin splitting of the $1D$ multiplet is predicted to be only a few MeV, so a single unresolved peak near 7030 MeV with a $\sim$ 260–320 MeV photon is the expected signature.

Falsifiable statements. The following observations would be incompatible with every potential fitted to $c\bar{c} + b\bar{b}$ in this work: (i) a $1P$ centroid outside 6745–6768 MeV, or any B_c excited state below 6690 MeV; (ii) a $B_c^*(2S) - B_c(2S)$ splitting outside 30–40 MeV (the present ATLAS+CMS value is 35.4 ± 2.4); (iii) a $1D$ centroid more than 30 MeV from 7035 MeV, or a $^3D_1$ heavier than the $^3D_3$; (iv) a $3\,^3S_1$ outside 7250–7340 MeV. Conversely, a measured $1P$ centroid determines the confining power α of Eq. (4) to ± 0.05 through Fig. 3, independently of the spectrum fit.

9 Conclusions

Fitting three families of potentials to the 2026 charmonium and bottomonium spectra and propagating them to $c\bar{b}$ without further input, we find:

1. The $c\bar{c} + b\bar{b}$ data prefer a nearly logarithmic confining potential, $\alpha \lesssim 0.1$, and the S-wave B_c levels are determined to ± 15 MeV independently of the potential; the P, D and $3S$ levels are not, and are the states to measure.

2. The hyperfine difference $\Delta_{1S} - \Delta_{2S}$ measured by CMS and LHCb is predicted correctly from $c\bar{c}+b\bar{b}$ and independently selects $\alpha \simeq 0.1$–0.2.

3. A fixed α_s overpredicts the $B_c^*(1S)$ hyperfine splitting measured by ATLAS by 10–20 MeV; a one-loop running coupling, with Λ fixed by $c\bar{c}$ and $b\bar{b}$, predicts $\Delta_{1S} = 63 \pm 15$ MeV and $\Delta_{2S} = 35 \pm 14$ MeV, in agreement with ATLAS, CMS and LHCb simultaneously, although it reproduces the $c\bar{c}$ and $b\bar{b}$ endpoints only at the 2σ level.

4. A self-consistent smeared contact term with a mass-independent range is excluded by the $b\bar{b}$ hyperfine splitting; the $b\bar{b}$ system is the arbiter between hyperfine prescriptions.

5. With a full uncertainty budget the excited B_c levels are predicted to ± 20–35 MeV, and the $1P$ and $1D$ multiplets have distinctive radiative signatures (Table 10).

6. The spin-averaged $B_c(2S)$ is predicted 11–24 MeV too high in every variant, at the edge of the budget.

The last point deserves comment. The $2S$ discrepancy survives changes of confining potential, of α_s treatment and of theory floor, and is several times the propagated uncertainty. It is the first level at which the first-order, non-relativistic treatment used here appears to fail for B_c while succeeding for $c\bar{c}$ and $b\bar{b}$, and we suggest that it reflects either relativistic corrections that do not interpolate linearly in $1/m$, or the proximity of the $2S$ to the BD threshold. A relativised treatment [7] or a coupled-channel calculation with the $c\bar{c} + b\bar{b}$-determined potential of this work would settle the question.

Table 10: Excited B_c states from the running-coupling Martin fit: mass with total uncertainty from Table 7 (MeV), $\langle v_c^2 \rangle$, and the dominant radiative lines as (final state, E_γ in MeV, Γ in keV, radiative branching fraction). Hadronic transitions are not included, so the branching fractions of the $2S$ and $3S$ states, which decay mainly through $\pi\pi$ emission, are of the radiative width only. Our $1P$, $1D$, $2P$ are the $2P$, $3D$, $3P$ of principal-quantum-number labelling.

State	Mass	$\langle v_c^2 \rangle$	Radiative lines (final, E_γ, Γ, BR)
$1\,^3P_0$	6699 ± 34	0.12	$1\,^3S_1$ (358, 59, 100%)
1^+	6734 ± 29	0.12	$1\,^3S_1$ (391, 71, 88%); $1\,^1S_0$ (450, 10, 12%)
$1^{+\prime}$	6746 ± 21	0.12	$1\,^1S_0$ (461, 116, 94%); $1\,^3S_1$ (402, 7, 6%)
$1\,^3P_2$	6760 ± 19	0.12	$1\,^3S_1$ (415, 92, 100%)
$2\,^1S_0$	6884 ± 19	0.31	$1^{+\prime}$ (137, 15, 89%); 1^+ (148, 2, 10%)
$2\,^3S_1$	6919 ± 17	0.31	$1\,^3P_2$ (157, 14, 41%); 1^+ (182, 12, 35%); $1\,^3P_0$ (216, 7, 21%)
$1\,^3D_1$	7025 ± 25	0.08	$1\,^3P_0$ (319, 73, 64%); 1^+ (285, 36, 32%); $1^{+\prime}$ (274, 3, 3%)
$1D_2$	7021 ± 11	0.08	$1^{+\prime}$ (270, 44, 53%); 1^+ (281, 31, 38%); $1\,^3P_2$ (257, 7, 9%)
$1D_2'$	7034 ± 16	0.08	1^+ (293, 43, 46%); $1^{+\prime}$ (282, 39, 42%); $1\,^3P_2$ (269, 11, 12%)
$1\,^3D_3$	7026 ± 15	0.08	$1\,^3P_2$ (261, 72, 100%)
$2\,^3P_0$	7106 ± 27	0.21	$2\,^3S_1$ (185, 24, 60%); $1\,^3S_1$ (733, 13, 32%); $1\,^3D_1$ (81, 3, 8%)
1^+	7132 ± 25	0.21	$2\,^3S_1$ (210, 30, 51%); $1\,^3S_1$ (756, 13, 21%); $2\,^1S_0$ (243, 8, 13%)
$1^{+\prime}$	7143 ± 23	0.21	$2\,^1S_0$ (254, 55, 62%); $1\,^1S_0$ (821, 16, 18%); $2\,^3S_1$ (221, 6, 7%)
$2\,^3P_2$	7153 ± 23	0.21	$2\,^3S_1$ (231, 47, 63%); $1\,^3S_1$ (775, 16, 21%); $1\,^3D_3$ (126, 10, 14%)
$3\,^1S_0$	7238 ± 34	0.32	$1^{+\prime}$ (94, 13, 78%); 1^+ (106, 3, 18%)
$3\,^3S_1$	7263 ± 12	0.32	$2\,^3P_2$ (109, 13, 38%); 1^+ (130, 12, 34%); $2\,^3P_0$ (155, 8, 22%)

Acknowledgments

This work grew out of the author's MSc project in the School of Physics and Astronomy, University of Glasgow, supervised by Christine Davies, whose guidance on the potential-model treatment of heavy quarkonia is gratefully acknowledged. The calculations are light enough to be reproduced in full on a single CPU core in under one hour; no high-performance computing resources were used. An AI assistant (Claude, Anthropic) was used to refactor the thesis code into the package described here, to implement the spin-dependent, fitting and radiative-transition modules, and to help draft the manuscript; the author takes full responsibility for the physics, the code and the results.

Code and data availability

The complete code, the fitted parameter sets and the scripts that regenerate every table and figure in this paper are available at `https://github.com/apoppong24/bc-spectrum-paper`; the experimental inputs are listed in Table 1.

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